

Multimodal Artificial Intelligence in Enterprise Robotic Process Automation

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Abstract—Digital transformation in enterprises has accelerated the adoption of Cognitive Robotic Process Automation (RPA) capable of processing unstructured and multimodal inputs such as documents, images, and speech. However, deploying end-to-end Large Language Models (LLMs) in high-frequency enterprise workflows introduces significant challenges related to prediction accuracy, computational latency, and operational cost. This paper proposes a modular two-stage framework for multimodal information extraction and entity resolution designed for domain-specific enterprise environments. The architecture separates signal interpretation from semantic reasoning. In the first stage, specialized perception modules convert multimodal inputs into textual representations. In the second stage, a unified Large Language Model performs zero-shot structured entity extraction from the transcribed content. To resolve extracted entities against large enterprise databases, the system employs a heterogeneous approach: merchant identities are resolved via robust lexical normalization and token-set fuzzy string matching, while complex product catalogs are mapped utilizing a hybrid retrieval mechanism that combines dense semantic representations from Sentence-BERT with sparse lexical matching via BM25. Experimental evaluation indicates that the proposed modular architecture provides an efficient, accurate, and scalable approach for integrating noisy multimodal inputs into structured enterprise data pipelines, reducing computational overhead while maintaining high extraction fidelity.

Index Terms—Multimodal Information Extraction, Robotic Process Automation, Large Language Models, Hybrid Retrieval

I. INTRODUCTION

Enterprise procurement workflows in emerging markets generate document inputs that are predominantly unstructured: handwritten order slips, photographed receipts, and verbal purchase requests transmitted over phone or messaging applications. In the Vietnamese retail and distribution sector, this pattern is especially pronounced. A significant share of small and medium enterprises (SMEs) record transactions on paper invoices and communicate orders verbally, often mixing Vietnamese with English product terminology and regional abbreviations [?]. Existing Robotic Process Automation (RPA) deployments handle these inputs through manual exception queues, which erode the efficiency gains that automation is intended to deliver.

Processing such inputs end-to-end with large language models (LLMs) is technically feasible but operationally impractical at scale. Invoking a multimodal LLM for every incoming document incurs per-token API costs and round-trip latency that,

across the high transaction volumes typical of distribution centres, accumulate to prohibitive expense [?]. At the same time, lightweight rule-based pipelines fail on the lexical variability introduced by tonal Vietnamese, bilingual product names, and inconsistent abbreviation conventions (e.g., *cty TNHH* for *công ty trách nhiệm hữu hạn*) [?]. Neither extreme satisfies the accuracy and throughput requirements of production RPA.

Prior work on document information extraction has focused on English-language, high-resource settings [?] or on isolated components—ASR for Vietnamese [?], [?], entity extraction via LLM prompting [?], and hybrid retrieval for product search [?], [?—without integrating these into a unified pipeline designed for enterprise-grade deployment constraints. No publicly available system addresses the combination of Vietnamese multimodal input, zero-shot entity extraction, and catalog-scale entity resolution under latency and cost budgets relevant to production RPA.

This paper describes a modular, two-stage pipeline that separates *signal transcription* from *semantic extraction*. In Stage 1, modality-specific modules—a cloud OCR service for document images and PhoWhisper [?] for spoken Vietnamese—convert raw inputs into a shared textual representation. In Stage 2, a single LLM performs zero-shot structured extraction of merchants, products, quantities, and units from the transcribed text. Extracted entities are then resolved against an enterprise catalog through a heterogeneous resolution layer: fuzzy string matching for merchant names and hybrid dense–sparse retrieval (Sentence-BERT + BM25, fused via Reciprocal Rank Fusion) for product SKUs, with an optional taxonomy-guided filter and a human-in-the-loop fallback for low-confidence cases.

The contributions of this paper are:

- 1) **System design:** A modular, eight-layer pipeline that processes Vietnamese visual and acoustic business documents into schema-compliant database records, with independent optimization of the extraction and resolution components.
- 2) **Heterogeneous entity resolution:** A domain-driven resolution strategy that applies fuzzy lexical matching to merchant names and hybrid dense–sparse retrieval to product descriptions, motivated by the distinct error characteristics of each entity type.

- 3) **Empirical evaluation:** Measurements of extraction accuracy (entity-level precision, recall, and F_1), product retrieval performance ($MRR@K$, $Hit@1$), ASR word error rate, and end-to-end throughput on a corpus of Vietnamese enterprise documents and audio recordings, with ablations across retrieval fusion strategies.
- 4) **Deployment analysis:** A cost and latency breakdown of the modular architecture compared to an end-to-end LLM baseline, informing practical deployment decisions for RPA integration.

II. RELATED WORK

A. Document Understanding and OCR-Based Information Extraction

Layout-aware document understanding has advanced considerably with the introduction of LayoutLM [?] and its successors, which jointly encode text tokens and their two-dimensional bounding-box coordinates using transformer self-attention. These models achieve strong performance on form understanding and document classification benchmarks, but require fine-tuning on annotated data in the target layout domain. End-to-end approaches such as Donut [?] eliminate the OCR stage entirely by learning to decode structured outputs directly from document images; however, they are computationally expensive at inference time and require substantial task-specific training data.

For enterprise RPA, where document formats vary across suppliers and input quality fluctuates between high-resolution scans and low-quality phone photographs, both categories of model face practical constraints. Our pipeline delegates visual understanding to a cloud OCR service with layout analysis and isolates the semantic extraction step in a downstream LLM, decoupling visual-quality sensitivity from extraction logic without requiring fine-tuning.

B. Vietnamese Automatic Speech Recognition

Vietnamese presents specific difficulties for ASR: it is a tonal, monosyllabic language with six tone classes, limited publicly available training data, and pronounced regional dialect variation. Wav2Vec 2.0 [?] demonstrated that self-supervised pre-training on unlabelled speech can significantly reduce the labelled data requirement for low-resource languages, including Vietnamese. The Zipformer architecture [?] further improved parameter efficiency by processing speech at multiple temporal resolutions.

PhoWhisper [?], developed by VinAI, fine-tunes the OpenAI Whisper encoder-decoder [?] on over 800 hours of multi-regional Vietnamese speech and achieves lower WER than Wav2Vec-based systems on conversational Vietnamese. VietASR [?] combines a HuBERT [?] encoder with a Zipformer decoder to maintain competitive accuracy while supporting real-time streaming. A limitation shared by both systems in the retail domain is that pre-training corpora do not cover the mixture of Vietnamese, English product names, and abbreviated trade terminology characteristic of spoken purchase orders.

This work selects PhoWhisper-base as the acoustic transcription module and applies an LLM post-correction layer specifically targeted at known phonetic substitution patterns in retail speech (e.g., the substitution of homophonic loanwords for technical product names).

C. Zero-Shot Information Extraction via LLMs

Traditional named entity recognition (NER) systems trained on annotated corpora—such as BiLSTM-CRF models [?]
—require substantial labelled data per entity schema and do not generalise when new entity types or document layouts are introduced. Prompt-based extraction using LLMs bypasses this requirement: given a schema description and a few illustrative examples, models such as GPT-4 and Gemini can extract structured records from free-form text in a zero-shot or few-shot setting [?].

The practical constraint is throughput. Invoking a frontier LLM for each document in a high-volume pipeline produces per-document latency in the range of one to several seconds and incurs API costs that scale linearly with token count. Our architecture limits LLM invocation to two narrow functions—entity extraction from transcribed text and optional taxonomy prediction—rather than using it for end-to-end document processing, thereby bounding both latency and cost.

D. Hybrid Retrieval for Entity Resolution

Sparse retrieval with BM25 [?] provides fast, exact-match recall but fails on synonyms and abbreviations. Dense retrieval with sentence embeddings [?] captures semantic similarity but can confuse closely related product specifications that differ only in a numeric attribute (e.g., LED 8W vs. LED 9W). Hybrid approaches that combine both signals through Reciprocal Rank Fusion [?] have been shown to outperform either strategy in isolation on heterogeneous retrieval benchmarks.

For Vietnamese product catalogs, which contain both semantically descriptive product names and opaque alphanumeric SKU codes, neither modality is universally preferable. We implement RRF as the primary fusion strategy within the Qdrant vector database [?], which supports parallel execution of dense and sparse retrieval in a single query. In addition, we evaluate two variants: (i) *Weighted Score Fusion* (WSF), which applies min-max normalisation before a linear combination of scores with tunable weight α , and (ii) *Weighted RRF* (WRRF), which applies α directly to the rank-based contributions, eliminating the normalisation requirement. The three strategies are compared empirically in Section ??.

A further distinction from prior retrieval work is the optional taxonomy-guided filter applied before retrieval. By conditioning the vector search on an LLM-predicted product category label, the effective catalog size presented to the retrieval engine is reduced, lowering the risk of cross-category semantic collisions in large-scale catalogs.

III. EXPERIMENTS AND RESULTS

We evaluate the performance of our proposed system on a dataset of enterprise documents and audio recordings.

IV. CONCLUSION

This paper presented a novel approach for integrating multimodal AI into enterprise RPA workflows, addressing the challenges of accuracy, latency, and cost.